Shawn Jain

San Francisco, CA
shawnjain.net

Summary

Researcher and engineer investigating core problems in open-ended learning, generative models, and autonomy. Experience spanning code generation, robotics, and natural language.

Education

Massachusetts Institute of Technology **September 2016 - September 2017**
M.Eng. Computer Science, A.I. Concentration Cambridge, MA

- Thesis: *VirtualHome: Learning to infer programs from synthetic videos of activities in the home*
- Research Area: Computer Vision; Advisor: Antonio Torralba

Massachusetts Institute of Technology **September 2012 - August 2016**
S.B. Electrical Engineering and Computer Science Cambridge, MA

- Best undergraduate lab project: *Automatic Projector Tilt Compensation System*, implemented on Xilinx FPGA

Illinois Mathematics and Science Academy **September 2010 - June 2012**
High School Diploma Aurora, IL

Experience

Zero Shot Fund **September 2025 - Present**
Founding General Partner (Part-Time) San Francisco, CA

- Raising a VC Fund for early stage AI startups. TechCrunch article.
- First investor in Foundry Robotics (marked up).

Synthefy **June 2024 - September 2025**
Co-Founder and CTO San Francisco, CA

- Raised a $6M seed round; hired and led a team of three researchers/engineers to develop SOTA time-series model.
- Secured first paid client: Algo8, an e-commerce analytics platform.
- Secured POC engagements: Samsung, Intel, and Daylight Computing.

OpenAI **January 2021 - October 2023**
Researcher (Member of Technical Staff) San Francisco, CA

- Code Generation: Taught GPT to iteratively debug code using a python interpreter. Released as Code Interpreter/Advanced Data Analysis in ChatGPT. Used by millions of paid subscribers.
- Open-Endedness: Algorithms that run forever and generate an endless variety of interesting artifacts. Co-Authored “Evolution through Large Models” [Paper], combining LLMs with these algorithms to create SOTA results across diverse domains. Pioneered this line of research combining Genetic Programming/Evolutionary Algos and LLMs.

Microsoft Research **September 2019 – September 2020**
AI Resident Redmond, WA

- “Do Transformers Understand Time?” [Blog] [Poster] Mentors: Hamid Palangi, Yonatan Bisk
- “Fast training and inference for NNs, applications to Transformer models.” Mentor: Greg Yang

Independent Researcher **February 2019 - August 2019**
Scholar Pittsburgh, PA

- Organized study groups with 3+ members, 2x per week; set agenda, kept engagement high for 12+ months.
- Implementations (most from scratch): Neural Networks/SGD, k-means clustering, SVM, GPs, Naïve Bayes, PCA/SVD applications, HOG features, decision trees. More at shawnjain.net

Uber Advanced Technologies Group **October 2017 – February 2019**
Perception Engineer Pittsburgh, PA

- Independently led research, prototyping, and production implementation of a learning algorithm to calibrate lidar intensity to the physical property of reflectance. US Patent 10,598,791 B2 [Patent]
- Delivered a turnkey calibration solution that works in a mixed lidar vendor fleet, including Velodyne HDL-64e
- The algorithm enabled an online lidar intensity-based localization system and an online lane extraction system

Writing, Code, and Demos

More at shawnjain.net

- Opportunities in Robotics Foundation Models [Blog]
- Evolution Through Large Models (ELM) [Paper]
- Do Transformers Understand Time? [Blog] [Poster]
- Reproducing Uber AI Labs' Deep Neuro-Evolution Paper [Blog] [Code]
- Wrappers vs. Runtimes [Blog]
- Gradient Descent and Chain Linked Systems [Blog] [Code]
- Object Detection Based on Lidar Intensity US Patent 10,598,791 B2 [Patent]
- VirtualHome: Learning to infer programs from synthetic videos of activities in the home [Thesis]
- Naive Bayes from scratch [Demo]
- NNs/SGD from scratch [Demo]
- SVM from scratch [Demo]
- Gaussian Processes from scratch [Demo]

Teaching and Talks

"The OpenAI Story" - The AI Podcast (ARD) — **Forthcoming**
Interviewee — Berlin, Germany

Synthetic Data - Forbes/MIT Imagination in Action — **Spring 2024**
Speaker — Cambridge, MA

Digital Electronics Lab, MIT 6.111 ("Digital Death") — **Fall 2016**
Teaching Assistant — Cambridge, MA

Signals and Systems, MIT 6.003 — **Spring 2017**
Teaching Assistant — Cambridge, MA

References

By Request